Lecture 2 - Probability and statistical learning theory

COMP3670/COMP6670 - Introduction to Machine Learning

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Recall the end of Lecture 1

- We set up the **supervised learning framework**: an unknown **target function** f, a dataset $\mathcal{D}_N$, a **hypothesis set** $\mathcal{H}$, an algorithm $\mathcal{A}$, and a learned hypothesis $\hat{h}$
- We saw a concrete instance, the **perceptron**, and an algorithm (PLA) that drives the **in-sample error** $E_{\text{in}}(\hat{h})$ to zero
- And then we hit a wall: on the inputs **outside** the dataset, the data gives us **no reason** to prefer one prediction over another
- We argued that learning is only feasible from a **probabilistic** point of view

The question we left open

Does a small $E_{\text{in}}(\hat{h})$ tell us anything about the error of $\hat{h}$ on **new, unseen** data, $E_{\text{out}}(\hat{h})$?

Why we cannot answer it yet

- Every word in that question is still **informal**:
  - What does "**new, unseen data**" mean?
  - What *is* the "**target function**" f, if the same input may lead to different outputs?
  - What does "**tells us something**" mean, if nothing is guaranteed?
- Each of these is answered by concepts from **probability and statistics**:
  - a **probability distribution** P generating the data
  - a **conditional distribution**, which says how the outcome depends in a probabilistic way on the features
  - a **probabilistic bound**, which quantifies "how likely" the generalisation gap $|E_{\text{out}}(\hat{h}) - E_{\text{in}}(\hat{h})|$ is small

Probability spaces and random variables

A random phenomenon

- In general, in processes considered within the supervised learning framework, the outcome is **not** completely determined by the input:
  - Two applicants with **identical** recorded features may repay or default
  - Two patients with the **same** test results may or may not have the disease
- This is not a defect of the data: the features simply do **not** contain all information about the outcome
- So the outcome is best described as **random**, and the whole data-generating process as a **random phenomenon**
- Probability theory is the mathematical language for random phenomena, and it rests on **three** objects: the **sample space**, the **collection of possible events** and a **probability measure**

The sample space and events

Definition (Sample space)

The **sample space** Ω is the set of *all* possible outcomes of a random phenomenon. A typical outcome is denoted $\omega \in \Omega$.

Example

- Toss a coin: $\Omega = \{H, T\}$
- Roll a die: $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Toss a coin twice: $\Omega = \{HH, HT, TH, TT\}$
- Draw a credit applicant from the population: $\Omega =$ the set of all applicants

Definition (Event)

An **event** is a subset $A \subset \Omega$. We say that A *occurs* if the realised outcome ω belongs to A.

Events are sets, so set operations have meanings

- Rolling a die, $\Omega = \{1, \ldots, 6\}$. Let $A = \{2, 4, 6\}$ ("even") and $B = \{5, 6\}$ ("at least 5")
- Then:
  - $A \cup B = \{2, 4, 5, 6\}$ is the event "A **or** B"
  - $A \cap B = \{6\}$ is the event "A **and** B"
  - $A^c = \{1, 3, 5\}$ is the event "**not** A"
  - $\emptyset$ is the **impossible** event, Ω is the **certain** event
- Two events are **disjoint** (or mutually exclusive) if $A \cap B = \emptyset$: they cannot occur together
  - $A = \{2, 4, 6\}$ ("even") and $B = \{1, 3, 5\}$ ("odd") are disjoint
- We denote by $\mathcal{S}$ the collection of all events we consider as *possible*; when Ω is finite we may simply take $\mathcal{S}$ to be **all** subsets of Ω
- The third object is a **probability measure** that associates a probability to each event

The probability measure

Definition (Probability measure)

A **probability measure** is a function $\mathbb{P}$ that assigns to each event $A \in \mathcal{S}$ a number $\mathbb{P}(A)$, and satisfies:

1. $0 \leq \mathbb{P}(A) \leq 1$ for every event A;
2. $\mathbb{P}(\Omega) = 1$;
3. if $A_1, A_2, \ldots$ are **pairwise disjoint** (i.e., $A_i \cap A_j = \emptyset$) events, then

$$\mathbb{P}\left(\bigcup_{i \geq 1} A_i\right) = \sum_{i \geq 1} \mathbb{P}(A_i).$$

- The triple $(\Omega, \mathcal{S}, \mathbb{P})$ is called a **probability space**
- These three rules, called **Kolmogorov's axioms**, are *all* we assume: everything else is derived

Reading the three axioms

- The axioms look abstract, but they are quite *natural*:
  - $0 \leq \mathbb{P}(A) \leq 1$: a probability is a number between 0 and 1. We often read it as a percentage: $\mathbb{P}(A) = 0.3$ means "the probability of A occurring is 30%"
  - $\mathbb{P}(\Omega) = 1$: *something* must happen. Rolling a die, the result is certainly one of $1, \ldots, 6$
  - $A_1 \cap A_2 = \emptyset \implies \mathbb{P}(A_1 \cup A_2) = \mathbb{P}(A_1) + \mathbb{P}(A_2)$: if two things cannot happen together, the probability that one *or* the other happens is the **sum** of their probabilities

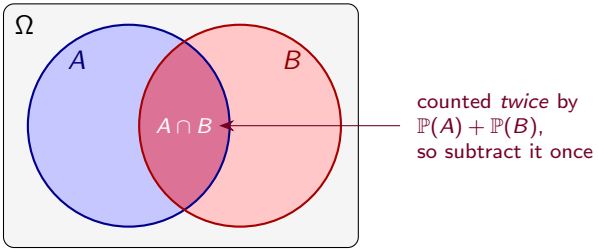
$$\mathbb{P}(\text{roll 2 or 4}) = \mathbb{P}(\{2\}) + \mathbb{P}(\{4\}) = \tfrac{1}{6} + \tfrac{1}{6} = \tfrac{1}{3}$$

The addition rule

- For *any* two events, disjoint or not,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \;\leq\; \mathbb{P}(A) + \mathbb{P}(B)$$

- Why subtract? Because adding $\mathbb{P}(A)$ and $\mathbb{P}(B)$ counts the **overlap twice**:



The addition rule: a practical example

Example

In a cohort, 60% of the students take calculus, 50% take linear algebra, and 30% take both. Pick a student at random. What is the probability they take *at least one* of the two?

- Let $A =$ "takes calculus" and $B =$ "takes linear algebra", so

$$\mathbb{P}(A) = 0.6, \qquad \mathbb{P}(B) = 0.5, \qquad \mathbb{P}(A \cap B) = 0.3$$

- Then

$$\mathbb{P}(A \cup B) = 0.6 + 0.5 - 0.3 = 0.8$$

- Note that $\mathbb{P}(A) + \mathbb{P}(B) = 1.1 > 1$: adding without subtracting the overlap gives nonsense

The union bound

- Iterating the addition rule gives, for any events $A_1, \ldots, A_m$,

$$\mathbb{P}\left(\bigcup_{i=1}^{m} A_i\right) \leq \sum_{i=1}^{m} \mathbb{P}(A_i)$$

- This is the **union bound**. It needs *no* assumption: the events may overlap in any way
- In words: **if none of m things is very likely, then it is not very likely that any of them happens**

Example

A system has 100 components, each failing on a given day with probability 0.0001. The probability that *some* component fails that day is at most $100 \times 0.0001 = 0.01$.

- Remember it. This will be important later.

From outcomes to numbers

- So far outcomes ω can be anything: heads/tails, an applicant, an image, a patient
- But our **data** is composed of **numbers**
- What we actually observe is never ω itself, but **measurements** taken on ω:
  - from an applicant ω: their income, years employed, debt
  - from a patient ω: blood pressure, cholesterol, and whether the disease is present
- A measurement is exactly a **function** from Ω to $\mathbb{R}$
- This is the concept of a **random variable**

Random variables

Definition (Random variable)

A **random variable** on a probability space $(\Omega, \mathcal{S}, \mathbb{P})$ is a function

$$X : \Omega \to \mathbb{R}$$

such that, for every $x \in \mathbb{R}$, the set $\{\omega \in \Omega : X(\omega) \leq x\}$ is an event, i.e., belongs to $\mathcal{S}$.

- The technical condition (called **measurability**) is there for one reason only: so that we are *allowed* to ask for the probability that $X \leq x$, and consequently that of other *events* of X
- Note that a random variable is **neither random nor a variable**: it is a deterministic function. The randomness is in *which* ω is drawn
- **Notation**: we write $\mathbb{P}(X \leq x)$ instead of the heavy $\mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x\})$, and similarly $\mathbb{P}(X \in B)$, $\mathbb{P}(X = x)$, and so on

A random variable: a worked example

Example

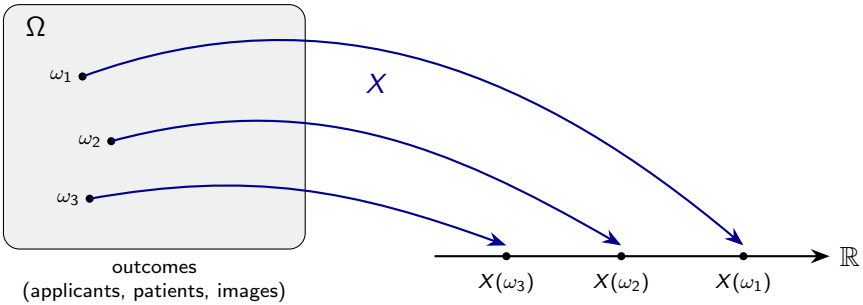
Toss a fair coin twice, so $\Omega = \{HH, HT, TH, TT\}$, and let $X =$ *the number of heads*.

ω	$X(\omega)$	$\mathbb{P}(\{\omega\})$
HH	2	$1/4$
HT	1	$1/4$
TH	1	$1/4$
TT	0	$1/4$

- X is a **rule**: it turns each outcome into a number, here one of $0, 1, 2$
- Two *different* outcomes may give the *same* number: $X(HT) = X(TH) = 1$

- The measurability condition in action, with $x = 1$: $\{\omega : X(\omega) \leq 1\} = \{HT, TH, TT\}$ is an event, so $\mathbb{P}(X \leq 1) = \frac{3}{4}$
- Similarly $\mathbb{P}(X = 1) = \mathbb{P}(\{HT, TH\}) = \frac{1}{2}$ and $\mathbb{P}(X = 2) = \frac{1}{4}$

Random variables: the picture



- $\mathbb{P}$ lives on Ω; X **transports** it to $\mathbb{R}$, where we can compute
- From now on we work entirely on the number line, and "ignore" Ω

Random variables on a population of patients

Example

Let Ω be a population of patients, each equally likely to be drawn, and define three random variables on it: $X_1 = \text{age}$, $X_2 = \text{blood pressure}$, $Y = 1$ if the disease is present and 0 otherwise.

patient ω	$X_1(\omega)$	$X_2(\omega)$	$Y(\omega)$
ω_1	41	128	0
ω_2	67	152	1
ω_3	55	139	1
ω_4	34	118	0
ω_5	72	161	1
ω_6	58	145	0

- Each **column** is a random variable; each **row** is one outcome ω, which the three measurements describe

Two types of random variable

- Which values a random variable can take splits them into **two** families, and the mathematics is different in each case

	Discrete	**Continuous**
values	separate values you can *list*	every value on a *scale*
examples	number of heads; spam / not spam	height; temperature; house price
$\mathbb{P}(X = x)$	can be positive	**always zero** for each single x
described by	a **probability mass function**	a **probability density function**
$\mathbb{P}$ **computed by**	**adding**	measuring **areas** under curves

- In supervised learning, **classification** is the case where Y is discrete, and **regression** the case where Y is continuous; the input vector can contain both discrete and continuous variables

Discrete random variables

Definition (Discrete random variable)

X is **discrete** if it takes values in a finite or countable set $\{x_1, x_2, \dots\}$. Its **probability mass function** (pmf) is

$$p_X(x) \coloneqq \mathbb{P}(X = x),$$

which satisfies $p_X(x) \geq 0$ and $\sum_{i \geq 1} p_X(x_i) = 1$.

- All probabilities are then obtained by **summing**:

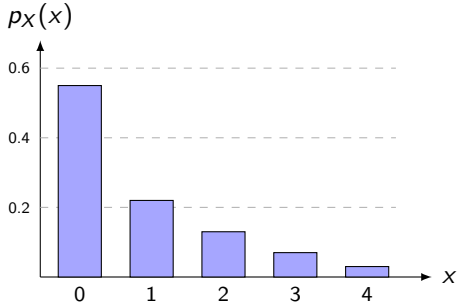
$$\mathbb{P}(X \in B) = \sum_{x_i \in B} p_X(x_i)$$

- In machine learning, discrete random variables model **labels**: classification is precisely the case in which Y is discrete

A discrete feature

Example

Draw an applicant at random and let X be the **number of late payments** recorded in the last two years, a possible *discrete input feature* of the credit approval problem of Lecture 1.



- $p_X(0) = 0.55$, $p_X(1) = 0.22$, $p_X(2) = 0.13$, $p_X(3) = 0.07$, $p_X(4) = 0.03$
- They add to 1, as they must
- "At least two late payments" is obtained by **adding the probabilities**:

$$\mathbb{P}(X \geq 2) = 0.13 + 0.07 + 0.03 = 0.23$$

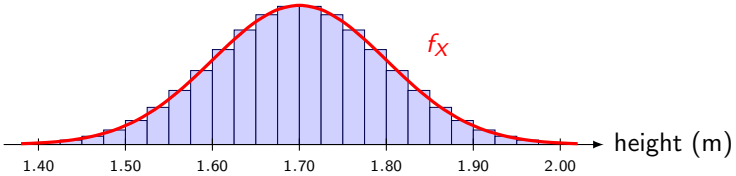
- Note that **inputs**, not only outputs, are random variables: this pmf describes the distribution of this random variable among the applicants

Continuous random variables

- Let X be the **height** of a randomly chosen adult, measured as precisely as we like
- What is $\mathbb{P}(X = 1.75 \text{ m})$? Not "someone about 1.75", but *exactly* $1.7500000\ldots$
- Nobody is *exactly* that tall, so the probability is **0**. And the same holds for *every* single value
- So listing $\mathbb{P}(X = x)$ value by value gives only zeros: a pmf carries **no information** here
- But these questions do have sensible answers:
  - $\mathbb{P}(1.70 \leq X \leq 1.80) \approx 0.30$
  - $\mathbb{P}(X \geq 1.90) \approx 0.02$
- For continuous quantities, probability attaches to **intervals**, not to single points

From a histogram to a curve

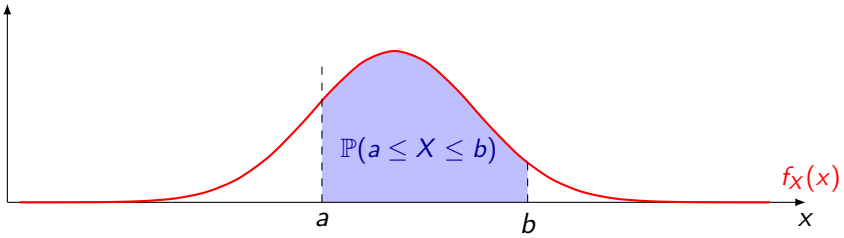
- Measure the heights of many people and draw a **histogram**: the **area** of each bar is the proportion of people in that range, so its height is that proportion *divided by* the width of the bin



- As the bins get **narrower**, the tops of the bars trace out a **curve**
- That limiting curve is the **probability density function** of X, written f_X

Density: probability is area

- In the histogram, the probability of a range was the **total area of the bars** in it. In the limit, it becomes the **area** under the curve



- The **area** under f_X between a and b is the probability that X falls in $[a, b]$
- The **total** area under f_X is 1, and the area over a single point is 0, as it must be
- Careful: $f_X(x)$ is **not** a probability. It is a *density*, and it may well exceed 1; only **areas** are probabilities

Continuous random variables: the definition

- The mathematical tool that measures "area under a curve" is the **integral**, written $\int$. With it, the definition reads:

Definition (Continuous random variable)

X is **continuous** if there exists a function $f_X : \mathbb{R} \to \mathbb{R}_+$, called its **probability density function** (pdf), such that

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)\,dx \qquad \text{for all } a \leq b.$$

Necessarily $f_X(x) \geq 0$ and $\displaystyle\int_{-\infty}^{\infty} f_X(x)\,dx = 1$.

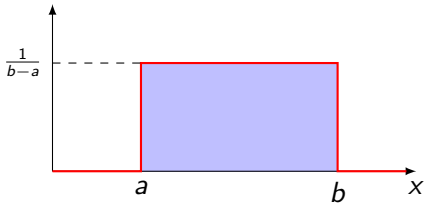
- For now, read $\displaystyle\int_a^b f_X(x)\,dx$ as nothing more than **"the area under f_X between a and b"**
- What that symbol *means*, and how to compute it, is the subject of the **next section**

Continuous example: the Uniform distribution

Example (Uniform $[a, b]$)

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

- "No region of $[a, b]$ is preferred": a random instant in a day, a random point on a segment
- Here we can check the total area by hand, since it is a **rectangle**: $(b-a) \cdot \frac{1}{b-a} = 1$
- $\mathbb{P}(X \leq x) = \dfrac{x-a}{b-a}$ for $x \in [a, b]$

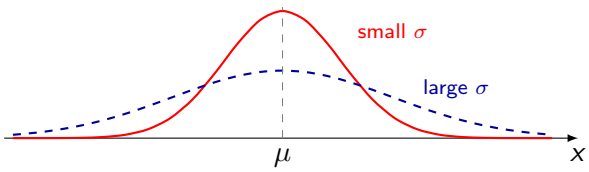


Continuous example: the Normal (Gaussian) distribution

Example (Normal(μ, σ^2))

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R},\ \mu \in \mathbb{R},\ \sigma > 0$$

- The "bell curve": heights, measurement errors, exam marks
- μ locates the centre, σ controls the spread, and it is symmetric around μ
- Do not be put off by the formula: all you need is the *shape*
- It is by far the most important continuous distribution



Random vectors

- A single number rarely suffices: an applicant is described by *several* features
- A **random vector** is a function $\mathbf{X} : \Omega \to \mathbb{R}^n$, $\mathbf{X} = (X_1, \ldots, X_n)$, each coordinate being a random variable
  - *e.g.*, $\mathbf{X} = (\text{income}, \text{years employed}, \text{late payments})$, so $n = 3$
- Exactly as for a single random variable, its **joint distribution** is described by a mass function or a density, now of n arguments:
  - all coordinates **discrete**: a joint pmf $p_{\mathbf{X}}(x_1, \ldots, x_n) = \mathbb{P}(X_1 = x_1, \ldots, X_n = x_n)$, and probabilities are obtained by **adding**
  - all coordinates **continuous**: a joint density $f_{\mathbf{X}}$, and probabilities are **volumes** under it, the n-dimensional version of "area under the curve"

Dependence and patterns

- Knowing the distribution of each coordinate X_i *separately* does **not** determine the joint distribution
- *e.g.*, knowing the distribution of people's heights, and separately of their weights, tells us nothing about **how the two go together**
- Yet that is precisely the useful information: tall people tend to be heavier, and it is this **dependence** that lets us predict weight from height

The point of the whole framework

Machine learning does not study the features and the outcome separately. It studies their **joint** distribution, because all the predictive information lives in the **dependence** between them.

The ML statistical model

The statistical model behind supervised learning

There is a probability space $(\Omega, \mathcal{S}, \mathbb{P})$ and a random vector

$$(\mathbf{X}, Y), \qquad \mathbf{X} \in \mathcal{X} \subset \mathbb{R}^n, \ Y \in \mathcal{Y},$$

with a **joint distribution** P that is **fixed but unknown**. The dataset is a *sample* of $(\mathbf{X}, Y)$.

- $\mathbf{X}$ is the feature vector, Y the outcome, *both* random
- P describes the phenomenon: how features occur, *and* how outcomes relate to them
- "**New, unseen data**" now has a precise meaning: a fresh draw of $(\mathbf{X}, Y)$ from the *same* P
- This single assumption is what makes generalisation possible at all; and it is also the assumption that fails when a model is deployed on a "population" different from the one it was trained on

Probability within the learning framework

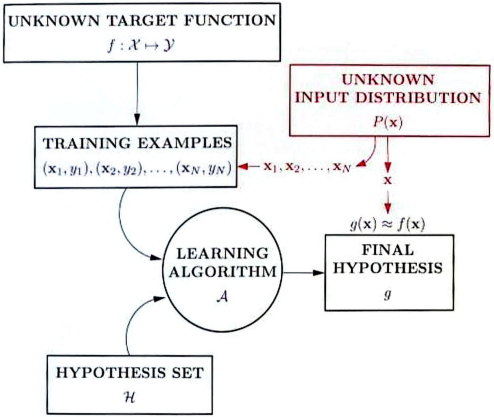


Figure 1.9: Probability added to the basic learning setup

Source: Abu-Mostafa, Y., Magdon-Ismail, M., & Lin, H.-T. (2012). *Learning from Data*. AMLBook. Figure 1.9.

Expectation, the integral and variance

Summarising a random variable by numbers

- The distribution of X contains **all** the information about X, but it is a whole function
- In practice we want a few **numbers** that summarise it:
  - where is X **centred**? $\rightarrow$ the **expectation**
  - how much does X **vary** around that centre? $\rightarrow$ the **variance**
- We start with the discrete case, where expectation is just a **weighted average**

Expectation of a discrete random variable

Definition (Expectation, discrete case)

Let X be a discrete random variable with values $x_1, x_2, \ldots$ and pmf p_X. Its **expectation** (or mean, or expected value) is

$$\mathbb{E}[X] := \sum_{i \geq 1} x_i \, p_X(x_i),$$

provided the sum is well defined.

- It is the average of the possible values, each **weighted** by how likely it is
- It is **not** the "most likely value", and it need not be a possible value at all

Example (Fair die)

$$\mathbb{E}[X] = 1 \cdot \tfrac{1}{6} + 2 \cdot \tfrac{1}{6} + 3 \cdot \tfrac{1}{6} + 4 \cdot \tfrac{1}{6} + 5 \cdot \tfrac{1}{6} + 6 \cdot \tfrac{1}{6} = \frac{21}{6} = 3.5$$

The expectation of an indicator

- Let A be an event and consider the random variable $\mathbb{1}\{A\}$, equal to 1 if A occurs and 0 otherwise

Probabilities are expectations in this case

$$\mathbb{E}\big[\mathbb{1}\{A\}\big] = 1 \cdot \mathbb{P}(A) + 0 \cdot (1 - \mathbb{P}(A)) = \mathbb{P}(A)$$

- This tiny identity is used constantly. For instance, the error of a classifier h,

$$\mathbb{P}\big(h(\mathbf{X}) \neq Y\big) = \mathbb{E}\big[\mathbb{1}\{h(\mathbf{X}) \neq Y\}\big],$$

  is simultaneously a **probability** and an **expectation**

Functions of a random variable and linearity

- If $g : \mathbb{R} \to \mathbb{R}$, then $g(X)$ is again a random variable, and in the discrete case

$$\mathbb{E}[g(X)] = \sum_{i \geq 1} g(x_i)\, p_X(x_i)$$

- *e.g.*, for a fair die and $g(x) = x^2$: $\mathbb{E}[X^2] = 1^2 \cdot \frac{1}{6} + 2^2 \cdot \frac{1}{6} + \cdots + 6^2 \cdot \frac{1}{6} = \frac{91}{6} \approx 15.17$
- **Note carefully**: in general $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$. In the example, $g(\mathbb{E}[X]) = 3.5^2 = 12.25 \neq 15.17$

Proposition (Linearity of expectation)

For random variables X, Y and constants $a, b \in \mathbb{R}$,

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y].$$

- Linearity is inherited from the linearity of sums and integrals

The continuous case needs the integral

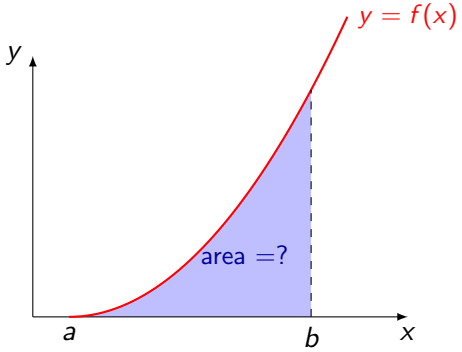
- For a continuous random variable, the natural definition of expectation replaces the sum by an **integral** and the pmf by the pdf:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx$$

- So before we can compute the expectation of a continuous quantity, we need to know what $\int$ **means** and how to evaluate it
- The next few slides present the basics of integral **calculus** and how to numerically compute integrals

The integral: area under a curve

- Let f be a function on an interval $[a, b]$. We want to give a meaning to the **area** between the graph of f and the x-axis



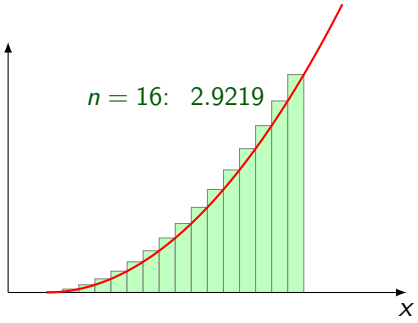
- For a rectangle or a triangle we know how to do this. For a curved region, we do not, so we **approximate**

Approximating by rectangles

- Split $[a, b]$ into n subintervals of equal width $\Delta x = \dfrac{b-a}{n}$, with endpoints

$$a = x_0 < x_1 < \cdots < x_n = b, \qquad x_i = a + i\,\Delta x$$

- On each subinterval, replace the curve by a **constant**: the value of f at a point $x_i^* \in [x_{i-1}, x_i]$



Riemann sums

Definition (Riemann sum)

Given a partition $a = x_0 < x_1 < \cdots < x_n = b$ and sample points $x_i^* \in [x_{i-1}, x_i]$, the **Riemann sum** of f is

$$R_n = \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i, \qquad \Delta x_i = x_i - x_{i-1}.$$

- Each term $f(x_i^*)\Delta x_i$ is the (signed) area of one rectangle
- Regions **above** the axis count as positive, regions **below** count as negative, so R_n approximates a **net** area
- In the picture: $R_4 = 3.75$, $R_8 = 3.1875$, $R_{16} = 2.9219$, and the numbers are decreasing towards something
- That "something" is the integral

The definite integral

Definition (Definite integral)

If the Riemann sums R_n approach a single number as the widths of all subintervals tend to 0, we say f is **integrable** on $[a, b]$ and call that number the **definite integral** of f from a to b:

$$\int_a^b f(x)\,dx := \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i.$$

- Vocabulary: a and b are the **limits of integration**, f is the **integrand**, and x is a **dummy variable** ($\int_a^b f(x)dx = \int_a^b f(t)dt$)
- Interpretation: $\int_a^b f(x)\,dx$ is the **net area** between the graph of f and the x-axis over $[a, b]$

Integrating over the whole line

- Densities are defined on *all* of $\mathbb{R}$, so we constantly need **infinite** limits of integration
- $\int_{-\infty}^{\infty} f(x)\,dx$ means exactly what it says: the **net area** under f over the whole line. It is obtained by letting the interval grow in both directions,

$$\int_{-\infty}^{\infty} f(x)\,dx \;=\; \lim_{a\to-\infty,\; b\to+\infty} \int_{a}^{b} f(x)\,dx$$

- Nothing changes in *how* it is computed: it is the **same** limit of Riemann sums, and the same rules apply
- An infinitely wide region may still have **finite** area, provided f decays fast enough as x grows. That is exactly what lets a density satisfy

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1$$

Properties of the definite integral

Proposition

Let f, g be integrable on $[a, b]$ and $c \in \mathbb{R}$. Then

$$\int_a^b \big[f(x) + g(x)\big]\,dx = \int_a^b f(x)dx + \int_a^b g(x)dx$$

$$\int_a^b c\,f(x)\,dx = c\int_a^b f(x)\,dx$$

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx, \qquad a \le c \le b$$

$$\text{if } f(x) \le g(x) \text{ on } [a,b], \text{ then } \int_a^b f(x)\,dx \le \int_a^b g(x)\,dx$$

- The first two say the integral is **linear**: this is exactly why expectation is linear

Exact areas and areas we can only approximate

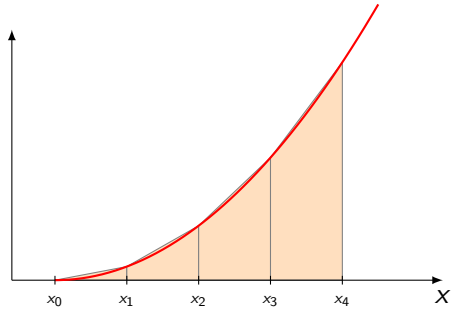
- For **some** functions the area under the curve is known **exactly**, by a formula:
  - the uniform density, because the region is a *rectangle*
  - $f(x) = x^2$ on $[0, 2]$, whose exact area is $\frac{8}{3} = 2.6667$, precisely the number our Riemann sums $3.75,\ 3.1875,\ 2.9219$ were approaching
- Calculus provides rules that produce such exact answers for many familiar functions, *e.g.* powers $f(x) = x^k$ for $k \in \mathbb{R}$ and the exponential function $f(x) = e^{cx}$ for $c \neq 0$, and for some combinations of them
- But for **most** functions no such formula exists. The normal probability density is an example:

$$\int_0^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx \ = \ ?$$

  It can be proved that **no formula** can give this area exactly
- The integral was *defined* as a limit of approximations, so we may go back to approximating, now **deliberately** and with a **guaranteed error**, in what is called **numerical integration**

Trapezoids instead of rectangles

- Approximate the area under the curve by **trapezoids** joining consecutive points of the graph by a **straight line segment**



- The trapezoid with base Δx and heights $f(x_{i-1})$, $f(x_i)$ has area $\dfrac{f(x_{i-1}) + f(x_i)}{2}\,\Delta x$

The Trapezoidal Rule

- Adding the areas of the n trapezoids, every interior height is counted **twice**:

Trapezoidal Rule

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)\Big]$$

with $\Delta x = \dfrac{b-a}{n}$ and $x_i = a + i\,\Delta x$.

Theorem (Error of the trapezoidal rule)

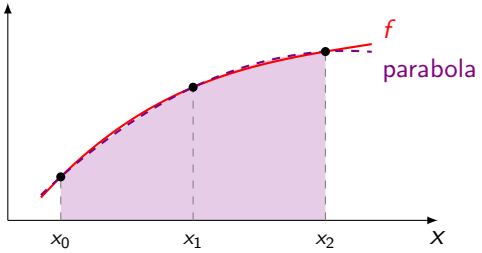
Let $E_T = \displaystyle\int_a^b f(x)\,dx - T_n$. *If* f *is well-behaved, there exists* $K_f > 0$ *such that*

$$|E_T| \leq \frac{K_f (b-a)^3}{12\,n^2}.$$

- Doubling n divides the error bound by 4: the rule is of **order** n^{-2}

Fitting parabolas instead of lines

- A straight line through two points is a crude local model. Through **three** equally spaced points there is exactly **one parabola**, and it approximates the curve much better



- The two curves **agree at the three points** and stay close in between, so the shaded area is a good approximation. Computing it with *known* integral computation, since it is the area under a polynomial, gives

$$\int_{x_0}^{x_2} f(x)\,dx \;\approx\; \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + f(x_2)\Big]$$

Simpson's Rule

- Apply the three-point rule to $[x_0, x_2]$, then to $[x_2, x_4]$, and so on, which requires n **even**:

Simpson's Rule (n even)

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big]$$

Theorem (Error of Simpson's rule)

Let $E_S = \int_a^b f(x)\,dx - S_n$. *If* f *is well-behaved, there exists* $C_f > 0$ *such that*

$$|E_S| \leq \frac{C_f (b-a)^5}{180\, n^4}.$$

- Order n^{-4} instead of n^{-2}: doubling n divides the error bound by 16. Note the coefficient pattern $1, 4, 2, 4, 2, \ldots, 4, 1$

Back to expectation: the continuous case

Definition (Expectation, continuous case)

Let X be continuous with pdf f_X. Then

$$\mathbb{E}[X] \coloneqq \int_{-\infty}^{\infty} x\, f_X(x)\, dx, \qquad \text{and} \qquad \mathbb{E}[g(X)] \coloneqq \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx.$$

- Same idea as before: each value x weighted by how likely it is, sums replaced by integrals
- **Notation used from now on.** To avoid writing the discrete and continuous cases separately, we write

$$\mathbb{E}[g(X)] = \int g(x)\, dP(x),$$

which means *a sum* when X is discrete and *an integral* when X is continuous.

Examples of continuous expectations

Example (Uniform $[a, b]$)

$$\mathbb{E}[X] = \int_a^b x \frac{1}{b-a}\, dx = \frac{a+b}{2}$$

which is the *midpoint* of the interval, as the symmetry of the density already suggests.

Example (Normal(μ, σ^2))

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}\, dx = \mu$$

by symmetry of the density around μ.

- But in general we do not have a closed formula and we need to approximate numerically

A note on integrals in many variables

- A feature vector $\mathbf{X} = (X_1, \ldots, X_n)$ has a joint density $f_{\mathbf{X}}$ of n arguments, so its expectations involve **one integral per variable**:

$$\mathbb{E}[g(\mathbf{X})] = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} g(x_1, \ldots, x_n)\, f_{\mathbf{X}}(x_1, \ldots, x_n)\, dx_1 \cdots dx_n$$

- The **idea is unchanged**. For $n = 2$ we chop the plane into small rectangles instead of an interval into small segments, multiply each by the height of f above it, and add: the area under a curve becomes the **volume** under a surface
- Writing this out is heavy, so **from now on we write a single integral sign**:

$$\mathbb{E}[g(\mathbf{X})] = \int g(\mathbf{x})\, dP(\mathbf{x})$$

- **Keep in mind what it stands for**: an n-fold sum when the coordinates are discrete, an n-fold integral when they are continuous, and a mixture of both when some coordinates are of each type

Variance and standard deviation

Definition (Variance)

The **variance** of a random variable X with $\mathbb{E}[X] = \mu$ is

$$\mathrm{Var}(X) := \mathbb{E}\big[(X - \mu)^2\big],$$

and the **standard deviation** is $\mathrm{sd}(X) = \sqrt{\mathrm{Var}(X)}$.

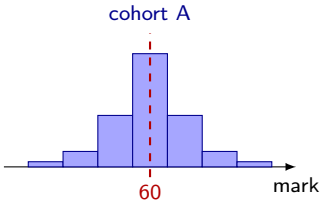
- It is the expected **squared** distance from the mean: a measure of **spread**. Squaring makes deviations above and below the mean both count as positive
- A useful reformulation, obtained by expanding the square and using linearity:

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2$$

The same mean, a very different variable

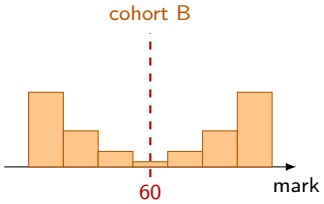
Example

Two cohorts sit the same exam, marked out of 100. Both have mean $\mathbb{E}[X] = 60$.



Marks concentrated around the mean

$\mathrm{Var}(X) = 124, \ \mathrm{sd}(X) \approx 11.1$



Marks split into two groups

$\mathrm{Var}(X) = 646, \ \mathrm{sd}(X) \approx 25.4$

- The **mean alone cannot tell the two apart**: in cohort B almost *nobody* scores near 60. The variance does, and it illustrates why a summary by the mean is incomplete

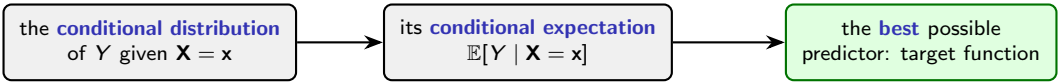
The plan from here

- **Conditioning** (next section) will describe how the outcome depends on the features
- **Loss and target function** will say what "best" means, and lead to the formal definition of the *target function*
- **Sampling** will formalise the concept of dataset and new, unseen data, and with them the in-sample and out-of-sample errors
- **Empirical Risk Minimisation** will fit the *learning algorithm* into this probabilistic framework, and leave us with the *generalisation gap* to control

Conditional distributions and conditional expectation

What prediction really is

- Prediction is not about the outcome Y *in general*: it is about the outcome Y **given** that we have observed the features $\mathbf{X} = \mathbf{x}$
- Knowing that an applicant earns \$52,000 does not tell us whether they will repay, but it **does** change how likely repayment is
- Observing $\mathbf{X} = \mathbf{x}$ restricts attention to a **subpopulation**, or an **event** in probabilistic terms, and within it the outcome has a different distribution
- The device that does this is **conditioning**:



The conditional distribution of the outcome

Definition (Conditional distribution)

Let $(\mathbf{X}, Y)$ be a random vector. The **conditional distribution of Y given $\mathbf{X} = \mathbf{x}$** is

$$p_{Y|\mathbf{X}}(y \mid \mathbf{x}) = \frac{p_{\mathbf{X},Y}(\mathbf{x}, y)}{p_{\mathbf{X}}(\mathbf{x})} \qquad \text{(discrete case, where } p_{\mathbf{X}}(\mathbf{x}) > 0\text{)}$$

$$f_{Y|\mathbf{X}}(y \mid \mathbf{x}) = \frac{f_{\mathbf{X},Y}(\mathbf{x}, y)}{f_{\mathbf{X}}(\mathbf{x})} \qquad \text{(continuous case, where } f_{\mathbf{X}}(\mathbf{x}) > 0\text{)}$$

- Read it as: **divide the joint by the marginal**. We restrict attention to the subpopulation with $\mathbf{X} = \mathbf{x}$, and renormalise so that its total probability is 1
- For each *fixed* $\mathbf{x}$ this is again a distribution in y: it is non-negative and adds (or integrates) to 1. Everything we know about distributions applies to it

Conditioning: a worked example

- Back to our six patients, with $X = \mathbb{1}\{\text{blood pressure} > 140\}$ and $Y = \mathbb{1}\{\text{disease present}\}$

ω	BP	X	Y
ω_1	128	0	0
ω_2	152	1	1
ω_3	139	0	1
ω_4	118	0	0
ω_5	161	1	1
ω_6	145	1	0

  - Without looking at X: $\mathbb{P}(Y = 1) = \frac{3}{6} = \frac{1}{2}$
  - Low blood pressure, $X = 0$: three patients, *one* diseased, so

$$p_{Y|X}(1 \mid 0) = \frac{\mathbb{P}(Y = 1, X = 0)}{\mathbb{P}(X = 0)} = \frac{1/6}{1/2} = \frac{1}{3}$$

  - High blood pressure, $X = 1$: three patients, *two* diseased, so

$$p_{Y|X}(1 \mid 1) = \frac{2/6}{1/2} = \frac{2}{3}$$

- Knowing X moves the probability from $\frac{1}{2}$ to $\frac{1}{3}$ or $\frac{2}{3}$: **that movement is the information the feature carries**

Conditional expectation

Definition (Conditional expectation)

The **conditional expectation of Y given $\mathbf{X} = \mathbf{x}$** is the expectation of Y under its conditional distribution:

$$\mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}] = \sum_{y} y \, p_{Y|\mathbf{X}}(y \mid \mathbf{x}) \qquad \text{or} \qquad \int y \, f_{Y|\mathbf{X}}(y \mid \mathbf{x}) \, dy.$$

- It is an ordinary expectation, computed with the *conditional* distribution in place of the unconditional one
- For each $\mathbf{x}$ this is a **number**; as $\mathbf{x}$ varies it defines a **function**

$$\eta(\mathbf{x}) \coloneqq \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}]$$

Conditional expectation of the patients

- In the patient example, Y takes only the values 0 and 1, so

$$\mathbb{E}[Y \mid X = 0] = 0 \cdot \tfrac{2}{3} + 1 \cdot \tfrac{1}{3} = \tfrac{1}{3}$$
$$\mathbb{E}[Y \mid X = 1] = 0 \cdot \tfrac{1}{3} + 1 \cdot \tfrac{2}{3} = \tfrac{2}{3}$$

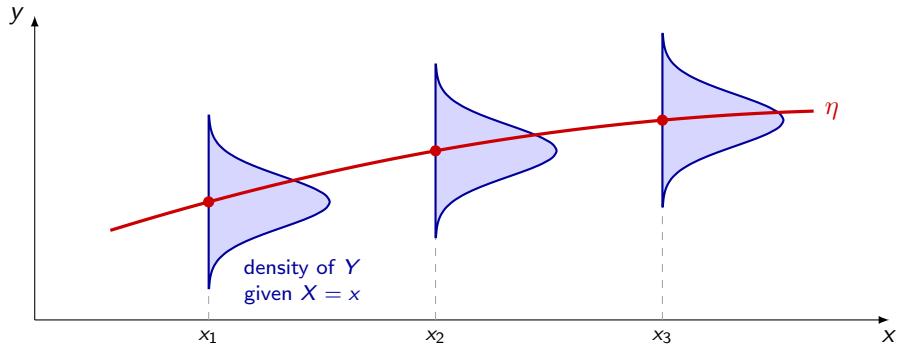
- For a **binary** outcome the conditional expectation *is* the conditional probability of $Y = 1$:

$$\mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}] = \mathbb{P}(Y = 1 \mid \mathbf{X} = \mathbf{x})$$

- So a single object, η, covers both settings: it is an **average outcome** when Y is numerical, and a **probability** when Y is a label

An illustration of conditional expectation

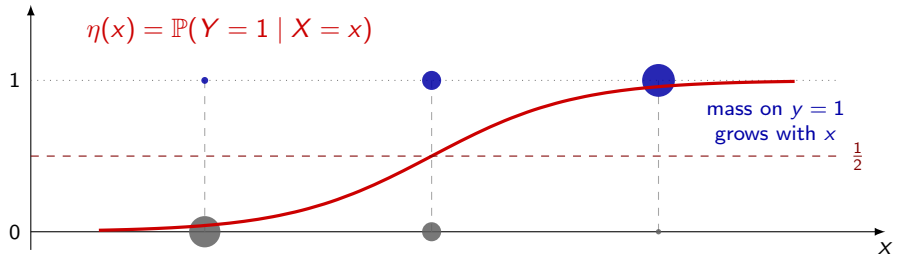
- At *each* value of x there is a whole **distribution** of Y. Its centre, as x varies, traces the function $\eta(x) = \mathbb{E}[Y \mid X = x]$



- Each bell curve is a *conditional* distribution: it integrates to 1 on its own. The **red dots** are its expectation, and the **red curve** joining them is η

An illustration of conditional expectation: the binary case

- When $\mathcal{Y} = \{0, 1\}$ the conditional distribution puts all its mass on **two** points, with probabilities $1 - \eta(x)$ on $y = 0$ and $\eta(x)$ on $y = 1$



- The **same** object: $\eta(x)$ is still the conditional expectation, and here it also *is* the conditional probability of $Y = 1$
- In the patient example this curve took just two values: $\eta(0) = \frac{1}{3}$ and $\eta(1) = \frac{2}{3}$

Loss function and target function

“Best” is meaningless until we say at what

- In Lecture 1 we assumed an **ideal rule** f that *best* maps every input to its output. We can now define what was missing: **best at what?**
- Two candidate rules will make different mistakes on different inputs. To compare them we must first decide **how much each mistake costs**
- That decision is a modelling choice, and it is made **before** any data is seen

Definition (Loss function)

A **loss function** ℓ assigns to each pair (observation, hypothesis) a non-negative number

$$\ell\big((\mathbf{x}, y), h\big) \;\geq\; 0,$$

the **price** of predicting $h(\mathbf{x})$ when the observed outcome is y.

The two losses we will use

- The choice of ℓ follows the **type** of the outcome, exactly as in the discrete/continuous split of the first section

Regression: squared loss

$$\ell\big((\mathbf{x}, y), h\big) = \big(y - h(\mathbf{x})\big)^2$$

Y is continuous, so we penalise *how far* the prediction is

Squaring makes errors on either side count the same, and penalises large errors much more

Classification: binary loss

$$\ell\big((\mathbf{x}, y), h\big) = \mathbb{1}\{h(\mathbf{x}) \neq y\}$$

Y is a label, so "how far" is meaningless: we only ask *right or wrong*

Pay 1 for a mistake, 0 for a correct prediction

Out-of-sample error

- The loss scores a hypothesis on **one** observation. To score it on the **phenomenon**, we average it over the distribution P that generates the data

Definition (Out-of-sample error)

The **out-of-sample error** of a hypothesis h is the expected loss under P,

$$L(h) \;=\; \mathbb{E}\big[\ell\big((\mathbf{X}, Y), h\big)\big] \;=\; \int \ell\big((\mathbf{x}, y), h\big)\, dP(\mathbf{x}, y).$$

- With the **squared loss**, $L(h) = \mathbb{E}\big[(h(\mathbf{X}) - Y)^2\big]$: the expected squared distance between y and $h(\mathbf{x})$ for a point $(\mathbf{x}, y)$ generated by P
- With the **binary loss**, $L(h) = \mathbb{E}\big[\mathbb{1}\{h(\mathbf{X}) \neq Y\}\big] = \mathbb{P}\big(h(\mathbf{X}) \neq Y\big)$: the probability of misclassifying a point $(\mathbf{x}, y)$ generated by P when using h

Target function

- The **target function** $f^\star$ is the rule with the smallest possible out-of-sample error

$$f^\star = \arg\min_h \; L(h), \qquad \text{the minimum taken over } \textbf{all} \text{ functions } h : \mathcal{X} \to \mathcal{Y}$$

Theorem

Let $\eta(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}]$. *Then the target function is*

$$\textbf{regression}, \textit{ squared loss}: \quad f^\star(\mathbf{x}) = \eta(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}]$$

$$\textbf{classification}, \textit{ binary loss}: \quad f^\star(\mathbf{x}) = \arg\max_{y \in \mathcal{Y}} \; p_{Y|\mathbf{X}}(y \mid \mathbf{x})$$

Proof: *Braga-Neto, U. M. (2024). Fundamentals of Pattern Recognition and Machine Learning. 2nd edition. Springer. Sections 2.3, 2.4 and 11.1*

- For a **binary** outcome $f^\star(\mathbf{x}) = \mathbb{1}\{\eta(\mathbf{x}) > \frac{1}{2}\}$: predict 1 when $\mathbb{P}(Y = 1 \mid \mathbf{X} = \mathbf{x}) > \frac{1}{2}$, that is, when $Y = 1$ is the more probable outcome given $\mathbf{X} = \mathbf{x}$, and predict 0 otherwise
- $f^\star$ is determined by P, and it is **unknown** precisely because P is unknown

Why this is the natural answer

- Conditioning lets us treat **one input at a time**: fix $\mathbf{X} = \mathbf{x}$, and the question becomes "given this single distribution of Y, which single value should we predict, when errors cost ℓ?"
- **Squared loss**: the number minimising the average squared distance to a distribution is its **mean**. So predict $\mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}] = \eta(\mathbf{x})$
- **Binary loss**: every wrong label costs the same 1, so we minimise the probability of being wrong by predicting the **most probable** label
- In the patient example, $\eta(1) = \frac{2}{3} > \frac{1}{2}$, so for high blood pressure the best possible prediction is "disease present": it is still **wrong one third of the time**, but it is **the best one can do!**

The target function takeaway

The "ideal rule $f^\star$" is determined by the loss we chose and the conditional distribution of Y given $\mathbf{X}$. Being the *best* does not make it *perfect*: in general $L(f^\star) > 0$.

The out-of-sample error of the target function

- We know *what* $f^\star$ is. How **wrong** is it?

Theorem

$$\textbf{regression}, \textit{ squared loss}: \quad L(f^\star) = \mathbb{E}\big[\mathrm{Var}(Y \mid \mathbf{X})\big]$$

$$\textbf{classification}, \textit{ binary loss}: \quad L(f^\star) = \mathbb{E}\big[\min\{\eta(\mathbf{X}), 1-\eta(\mathbf{X})\}\big]$$

Proof: *Braga-Neto, U. M. (2024). Fundamentals of Pattern Recognition and Machine Learning. 2nd edition. Springer. Sections 2.3, 2.4 and 11.1*

- Both average **over the features** how much Y still varies once $\mathbf{X}$ is known: the spread of Y left inside the subpopulation $\mathbf{X} = \mathbf{x}$, and the probability of the *less* probable label
- This error is **irreducible**: no hypothesis set, no algorithm and no sample size can go below it, and it is 0 **only** if the features determine the outcome

The out-of-sample error of the target function: example

- Our six patients again, using **only** $X = \mathbb{1}\{\text{blood pressure} > 140\}$

ω	BP	X	Y
ω_1	128	0	0
ω_2	152	1	1
ω_3	139	0	1
ω_4	118	0	0
ω_5	161	1	1
ω_6	145	1	0

- **1. Condition on each x.** Low BP ($X = 0$): $\omega_1, \omega_3, \omega_4$, one diseased, so $\eta(0) = \frac{1}{3}$. High BP ($X = 1$): $\omega_2, \omega_5, \omega_6$, two diseased, so $\eta(1) = \frac{2}{3}$. Each half of the population: $\mathbb{P}(X = 0) = \mathbb{P}(X = 1) = \frac{1}{2}$
- **2. The target function** predicts the more probable label: $f^\star(0) = \mathbb{1}\{\frac{1}{3} > \frac{1}{2}\} = 0$ and $f^\star(1) = \mathbb{1}\{\frac{2}{3} > \frac{1}{2}\} = 1$
- **3. The error left at each x** is the probability of the label it gave up, $\min\{\eta, 1-\eta\}$: it is $\frac{1}{3}$ at $x = 0$ and $\frac{1}{3}$ at $x = 1$
- **4. Average over the features**: $L(f^\star) = \frac{1}{2} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{3}$

- **Check by hand**: $f^\star$ says "healthy" for $\omega_1, \omega_3, \omega_4$ and "diseased" for $\omega_2, \omega_5, \omega_6$, so it is wrong on ω_3 and ω_6 only: $\frac{2}{6} = \frac{1}{3}$. **Blood pressure alone cannot do better**

Sampling

Population and sample

- The distribution P describes an entire **population** (all applicants, all emails, all patients), and it is unknown
- What we can obtain is a **sample**: finitely many observations drawn from that population
- Everything we can ever say about P, we must say through the sample. Statistics is the discipline that makes this passage rigorous
- But not every sample is informative. Compare:
  - asking 1000 people chosen at random about their income
  - asking 1000 people leaving a luxury car dealership
  - asking 1 person 1000 times
- The assumption that makes a sample informative is that of a **simple random sample**

Independent random variables

- Before defining a sample, we need the one assumption that makes a sample *informative*: **independence**. It is stated in terms of the random variables themselves

Definition (Independent random variables)

Random variables X and Y are **independent** if their joint distribution **factorises**:

$$p_{X,Y}(x,y) = p_X(x)\,p_Y(y) \qquad \text{(discrete case)}$$
$$f_{X,Y}(x,y) = f_X(x)\,f_Y(y) \qquad \text{(continuous case)}$$

for all x, y. Random variables $X_1, \ldots, X_N$ are independent if the joint mass function factorises as $p_{X_1}(x_1) \cdots p_{X_N}(x_N)$, and likewise for densities.

- When the variables are independent,

$$p_{Y|X}(y \mid x) = \frac{p_{X,Y}(x,y)}{p_X(x)} = \frac{p_X(x)\,p_Y(y)}{p_X(x)} = p_Y(y)$$

so knowing X **does not** change the distribution of Y

Independence: across observations, not across features

Example

Draw *two* patients at random from a large population, and record the blood pressure of each: X_1 for the first, X_2 for the second.

- X_1 and X_2 are **independent**: knowing that the first patient had a blood pressure of 161 tells us **nothing** about the second
- So probabilities **multiply**. If 10% of the population has the disease, then for two independently drawn patients

$$\mathbb{P}(\text{both have the disease}) = 0.1 \times 0.1 = 0.01$$

- Contrast this with *two measurements on the same patient*: age and blood pressure are **not** independent, since older patients tend to have higher blood pressure

Independence: what we assume, and what we exploit

- Note what would follow if the *features* were independent of the outcome: knowing the features would not change the probability of the outcome, and they would be **useless** for prediction
- So dependence between $\mathbf{X}$ and Y is not a nuisance: it is **the entire reason** prediction is possible

Two roles of independence

Independence **across observations** is what we **assume**, so that a sample carries new information at every example. Dependence **across features and outcome** is what we **exploit**, and it is what a model learns.

Simple random sample

- Each observation is a **pair** $(\mathbf{X}, Y)$: the features *and* the outcome, drawn together

Definition (Simple random sample)

A **simple random sample** of size N from a distribution P is a collection of random vectors

$$(\mathbf{X}_1, Y_1),\ (\mathbf{X}_2, Y_2),\ \ldots,\ (\mathbf{X}_N, Y_N)$$

that are

- **independent**, and
- **identically distributed**, each with distribution P.

We abbreviate this as **i.i.d.** and write $\mathcal{D}_N = \{(\mathbf{X}_1, Y_1), \ldots, (\mathbf{X}_N, Y_N)\}$.

- **Identically distributed**: every observation comes from the *same* "population"
- **Independent**: each observation brings *new* information, unrelated with the previous ones

Where the i.i.d. assumption fails

- **fine**: 1000 patients drawn at random from a hospital's records
- **not fine**: 1000 emails that are all replies in the same thread, so each one is highly predictable from the previous
- **not fine**: measurements taken on the same patient on 1000 consecutive days
- **not fine**: half of the sample is from last year's customers, before the company completely changed its marketing strategy, and half is from this year's. Behaviour should have shifted due to the marketing overhaul, so the observations are no longer *identically* distributed
- The theoretical generalisation guarantees of Statistical Learning Theory depend on the i.i.d. assumption, so it is imperative to have a sample (dataset) that satisfies it

The test point and generalisation

- The sample $\mathcal{D}_N$ is what we **learn from**. But the hypothesis will be used on observations that are **not** in it
- "New, unseen data" is now a precise object: a **test point**

$$(\mathbf{X}, Y) \sim P, \qquad \text{drawn } \textbf{independently} \text{ of } (\mathbf{X}_1, Y_1), \ldots, (\mathbf{X}_N, Y_N)$$

- Two requirements:
  - **same** P: the applicant we must decide on comes from the same population as those we learnt from
  - **independent**: the test point was in no way used to choose the hypothesis
- The credit approval problem of Lecture 1 is exactly this: the bank fits $\hat{h}$ on past applicants, and applies it to a *future* applicant, a fresh draw from P

In-sample and out-of-sample error

Definition (In-sample and out-of-sample error)

Given a loss ℓ and a hypothesis h,

$$\textbf{out-of-sample error}: \quad E_{\text{out}}(h) = L(h) = \mathbb{E}\big[\ell\big((\mathbf{X}, Y), h\big)\big], \quad (\mathbf{X}, Y) \text{ the test point}$$

$$\textbf{in-sample error}: \quad E_{\text{in}}(h) = L_{\mathcal{D}_N}(h) = \frac{1}{N}\sum_{i=1}^{N} \ell\big((\mathbf{X}_i, Y_i), h\big), \quad \mathcal{D}_N \text{ a sample}$$

- $E_{\text{in}}(h)$ is **RANDOM** because it depends on the sample that is observed; $E_{\text{out}}(h)$ is a **NUMBER**, unknown to us, but fixed
- With the binary loss this is exactly the quantity of Lecture 1:

$$E_{\text{in}}(h) = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\{h(\mathbf{X}_i) \neq Y_i\}, \qquad E_{\text{out}}(h) = \mathbb{P}\big(h(\mathbf{X}) \neq Y\big)$$

Empirical Risk Minimisation

What the learning algorithm can and cannot do

- We know what we **want**: the target function $f^\star$, the rule of smallest out-of-sample error
- We cannot have it, since the out-of-sample error is an expectation under the **unknown** P, so we cannot evaluate it, let alone minimise it
- What we *do* have is the sample, and E_{in}, which is computable
- So we make the only substitution available: **replace the expectation we cannot compute by the average we can**
- The result is a hypothesis $\hat{h} = \hat{h}(\mathcal{D}_N)$ built *from* the sample. Like any quantity computed from a sample, it is itself **random**: a different sample gives a different $\hat{h}$

Empirical risk minimisation

- This substitution *is* the learning algorithm $\mathcal{A}$ of Lecture 1. Let us name it

Empirical Risk Minimisation (ERM)

Given a hypothesis set $\mathcal{H}$ and a loss function $\ell : (\mathcal{X} \times \mathcal{Y}) \times \mathcal{H} \to \mathbb{R}_+$, define

$$\textbf{risk} \text{ (out-of-sample error)}: \quad L(h) = \mathbb{E}\big[\ell\big((\mathbf{X}, Y), h\big)\big] = \int \ell\big((\mathbf{x}, y), h\big)\, dP(\mathbf{x}, y)$$

$$\textbf{empirical risk} \text{ (in-sample error)}: \quad L_{\mathcal{D}_N}(h) = \frac{1}{N} \sum_{i=1}^{N} \ell\big((\mathbf{X}_i, Y_i), h\big)$$

and estimate a minimiser of L by a minimiser of $L_{\mathcal{D}_N}$:

$$\hat{h}_N = \arg\min_{h \in \mathcal{H}} \; L_{\mathcal{D}_N}(h).$$

- $L = E_{\text{out}}$ is what we **care** about but cannot compute; $L_{\mathcal{D}_N} = E_{\text{in}}$ is what we **can** compute

But does ERM work?

- ERM minimises $L_{\mathcal{D}_N}$, whereas what we want small is L
- For a **fixed** h, $L_{\mathcal{D}_N}(h)$ averages N i.i.d. quantities of expectation exactly $L(h)$, so for large N the two should be close
- But $\hat{h}_N$ is **not** fixed: it was **chosen using the very same sample** on which the error is measured
- Recall Lecture 1: the PLA drove E_{in} to zero *by construction*. A flexible enough $\mathcal{H}$ can drive $L_{\mathcal{D}_N}$ to zero for *any* sample, and this tells us nothing about L
- So the question is not whether averages converge to expectations, but whether they do so **uniformly over the whole of** $\mathcal{H}$, that is, for every hypothesis in $\mathcal{H}$ at the same time

The central question of Statistical Learning Theory

Under what conditions on $\mathcal{H}$ does a small empirical risk imply a small risk? We are once more at the **problem of induction**, but now with a formalised mathematical framework

Generalisation gap

Definition (Generalisation gap)

The **generalisation gap** of a hypothesis h on the sample $\mathcal{D}_N$ is

$$\left|E_{\text{out}}(h) - E_{\text{in}}(h)\right|$$

- Learning succeeds when **both** are small: a small in-sample error *and* a small gap. A small E_{in} on its own means nothing; it means something only once we can be confident that E_{out} is close to it
- The gap is **random**: E_{in} depends on the sample we happened to observe. So we cannot ask it to be small *always* since an unlucky sample is always possible
- What we can ask about is its **distribution**. Two summaries:
  - on average: $\mathbb{E}\big[\max_{h\in\mathcal{H}} \left|E_{\text{out}}(h) - E_{\text{in}}(h)\right|\big]$
  - or, more useful, the **probability of a large gap**: $\mathbb{P}\Big(\max_{h\in\mathcal{H}} \left|E_{\text{out}}(h) - E_{\text{in}}(h)\right| > \epsilon\Big)$

What we want to control

- The goal is, for example, to obtain

$$\mathbb{P}\Big(\max_{h \in \mathcal{H}} \big| E_{\text{out}}(h) - E_{\text{in}}(h) \big| > \epsilon \Big) \;\leq\; \delta$$

- Read it as: **with probability at least $1-\delta$**, no hypothesis in $\mathcal{H}$ has a gap larger than ϵ, and in particular

$$E_{\text{out}}(\hat{h}_N) \;\leq\; \underbrace{E_{\text{in}}(\hat{h}_N)}_{\text{computable}} + \epsilon$$

- ϵ is **how wrong** we may be, δ is **how often** we may be that wrong. Both must be controlled by things we can choose: the sample size N and the hypothesis set $\mathcal{H}$

Hume's problem of induction, answered probabilistically

We still cannot *prove* that a rule fitted on a finite sample will work on new data: Hume was right, and no bound changes that. What we can do is **bound the probability of being wrong, and by how much**. Induction is not made certain, it is made **quantifiable**.

The Statistical Learning framework

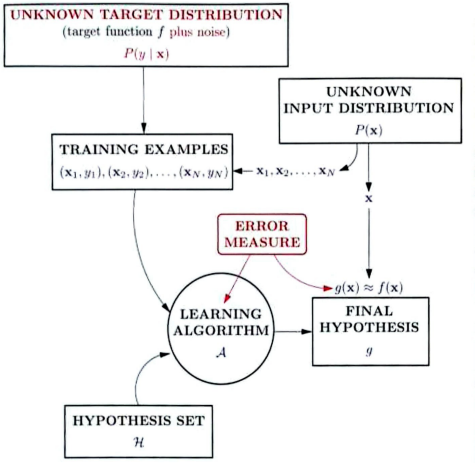


Figure 1.11: The general (supervised) learning problem

Source: Abu-Mostafa, Y., Magdon-Ismail, M., & Lin, H.-T. (2012). *Learning from Data*. AMLBook. Figure 1.11.

Next steps

- We have **formalised** the framework: a distribution P, a hypothesis set $\mathcal{H}$, a loss ℓ, the errors E_{in} and E_{out}, the algorithm as ERM, and the **generalisation gap** we must control
- Note what we have **not** done: we have not bounded that gap for a single concrete problem. The framework is built; the results are not yet in it
- **Next**, we ask which quantities actually **control** the gap, and what we pay for restricting to $\mathcal{H}$ (the **approximation error**):
  - the **bias-variance trade-off**, and **overfitting**, promised since Lecture 1
  - **PAC learning**: when is a gap of at most ϵ, with probability at least $1 - \delta$, achievable at all?
- **Then** the practical question: E_{out} is unknown, so how do we *estimate* it?
  - the **training**, **validation** and **test** split
  - **model selection**: choosing $\mathcal{H}$ itself from data, without corrupting the estimate

Lecture highlights

- **Probability**: probability spaces; a **random variable** transports probability from Ω to $\mathbb{R}$, described by a pmf (discrete) or a pdf (continuous)
- **Expectation and variance**: $\mathbb{E}[X] = \sum x\, p(x)$ or $\int x f(x) dx$; the integral is the **area under a curve**, computed exactly when a formula exists and otherwise by **numerical integration**
- **Conditioning**: the conditional distribution of Y given $\mathbf{X} = \mathbf{x}$ is the joint divided by the marginal, and its **conditional expectation** $\eta(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}]$ is a function of the features
- **Loss and target function**: choosing ℓ is choosing what "best" means; under the **squared loss** the target function is η, and under the **binary loss** it is the most probable label
- **Sampling**: the dataset is a **simple random sample** (i.i.d.); a **test point** is an independent draw from the same P, which is what makes E_{out} meaningful

Lecture highlights

- **The two errors**: $E_{\text{in}}(h)$ is an average over the sample, **random** but computable; $E_{\text{out}}(h)$ is an expectation over a **test point**, fixed but unknown
- **Empirical risk minimisation**: the learning algorithm of Lecture 1, written precisely as $\hat{h}_N = \arg\min_{h \in \mathcal{H}} E_{\text{in}}(h)$
- **Generalisation gap**: $|E_{\text{out}}(h) - E_{\text{in}}(h)|$. Learning succeeds only when a small E_{in} comes *together* with a small gap; a small E_{in} alone means nothing
- **What must be controlled**: since $\hat{h}_N$ is chosen *using* the sample that then measures it, we need $\mathbb{P}\big(\max_{h \in \mathcal{H}} |E_{\text{out}}(h) - E_{\text{in}}(h)| > \epsilon \big) \leq \delta$. Induction is not made certain, but **quantifiable**: a bound on how wrong (ϵ) and how often (δ)

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